Date: ___/___/___

# Web Scraping

- Technique for extracting data from webpages.
- Can be done in most languages.
- Purpose: Extracting info for analysis / research / other applications.

Why web scraping?

1. Data Collection.
2. Automation
3. Research

Libraries to use:

1. Beautiful Soup → XML, HTML : Structured Docs
2. lxml → Static / Poorly Structured
3. Selenium → Javascript: Dynamic Websites / Mimics Humans.
4. re
5. Scrapy

Rules:

1. Some prohibit scraping
2. Robots.txt should be checked
3. Dont overload servers
4. Avoid personal data
5. Use official APIs

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Code :

```
pip install selenium
pip install webdriver-manager

from selenium import webdriver
from selenium.webdriver.chrome.options import Options
from selenium.webdriver.common.keys import Keys
                .common.by import By
                .common.exceptions import NoSuchElementException
" " . " .chrome.service import Service
from webdriver_manager.chrome import ChromeDriverManager

driver = webdriver.Chrome(service=ChromeService(ChromeDriverManager()
    .install()))

url = 'https://www.daraz.pk/'
driver.get(url)

search_bar = driver.find_element(By.XPATH,'/html/body/div[1]')
    ....
search_bar.send_key('Samsung') # Types Samsung
                + Keys.RETURN
```

Assumes 1 if not given → one based index

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```
ad_grid = driver.find_element (By.XPATH, '/html/body')
ad_cards = ad_grid.find_elements (By.CLASS_NAME, '17m
for i in ad_cards:

    print (i.text)
```

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